

LANDSLIDE HAZARD MAPPING

Rudraprayag, Uttarakhand

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Research Framework & Methodology

Landslides are a major natural hazard in hilly regions, caused by the downslope movement of soil and rock. This study uses GIS-based analysis to identify hazard-prone areas and assess infrastructure risk in Rudraprayag.

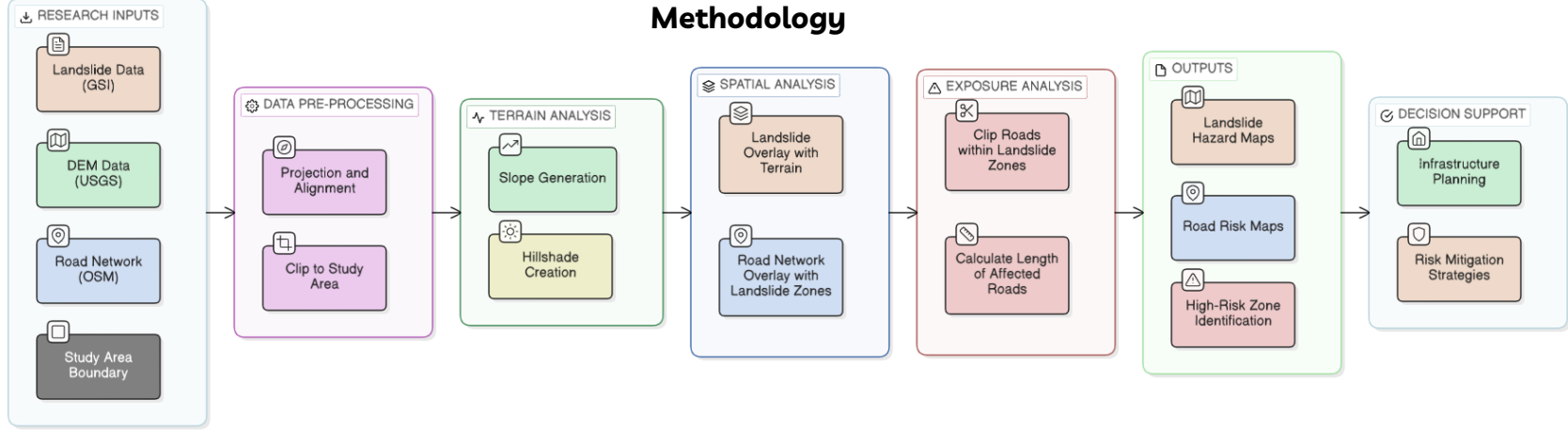
Aim

To assess and map **landslide hazard susceptibility** in Rudraprayag district, Uttarakhand, using **GIS-based spatial analysis**, and to evaluate the exposure of transportation infrastructure (road networks) to **landslide-prone zones** for risk-informed planning and mitigation.

Objective

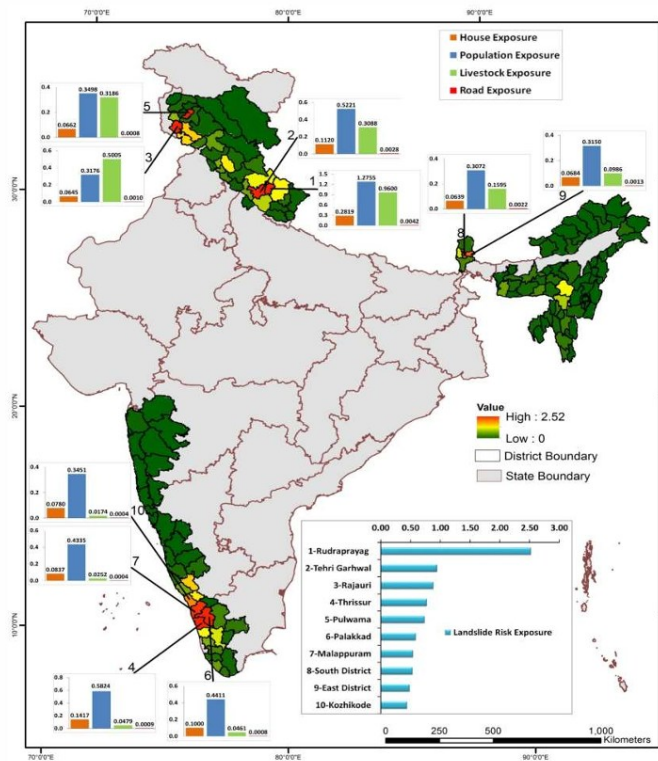
- To analyse terrain characteristics (slope and hillshade) influencing landslide occurrence.
- To map and delineate landslide-prone zones using GSI datasets.
- To integrate DEM-based terrain analysis with landslide inventory data.
- To assess the spatial relationship between road networks and landslide zones.
- To quantify the extent of **road infrastructure** exposed to **landslide hazards**.
- To identify high-risk corridors for planning and mitigation strategies.

Methodology



STUDY AREA – RUDRAPRAYAG DISTRICT, UTTARAKHAND

Landslide Risk Distribution in India



Source: Landslide Atlas of India, National Remote Sensing Centre, ISRO, Hyderabad

This study focuses on assessing landslide hazard and infrastructure exposure in a highly vulnerable Himalayan region.

Rudraprayag district in Uttarakhand is identified as a **high landslide susceptibility zone** due to its steep terrain, fragile geological formations, and frequent extreme rainfall events. According to the **Landslide Atlas of India**, the region exhibits significant exposure to landslide hazards. Given this vulnerability, the study area is selected to analyse landslide-prone zones and assess the risk to transportation infrastructure, particularly road networks.



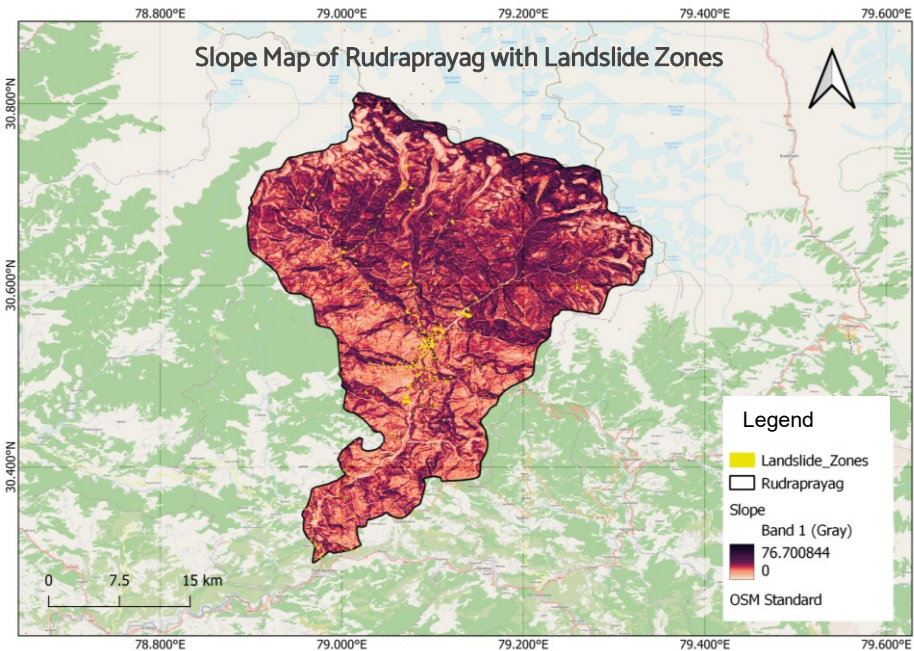
Why Rudraprayag?

- High landslide frequency
- Steep Himalayan terrain
- Critical road connectivity (pilgrimage routes)
- High infrastructure vulnerability

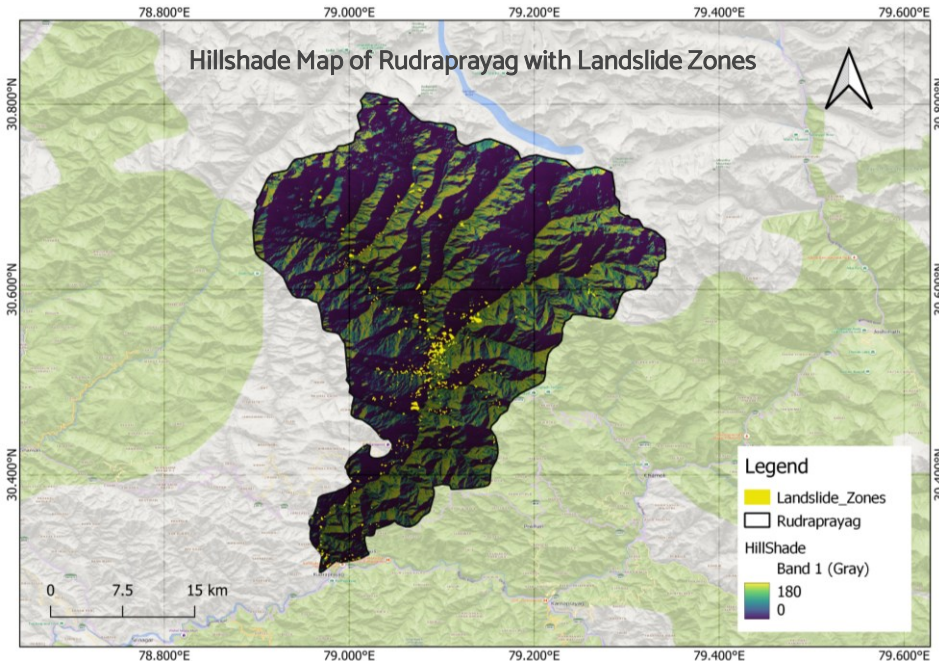


Terrain Analysis: Slope and Hillshade

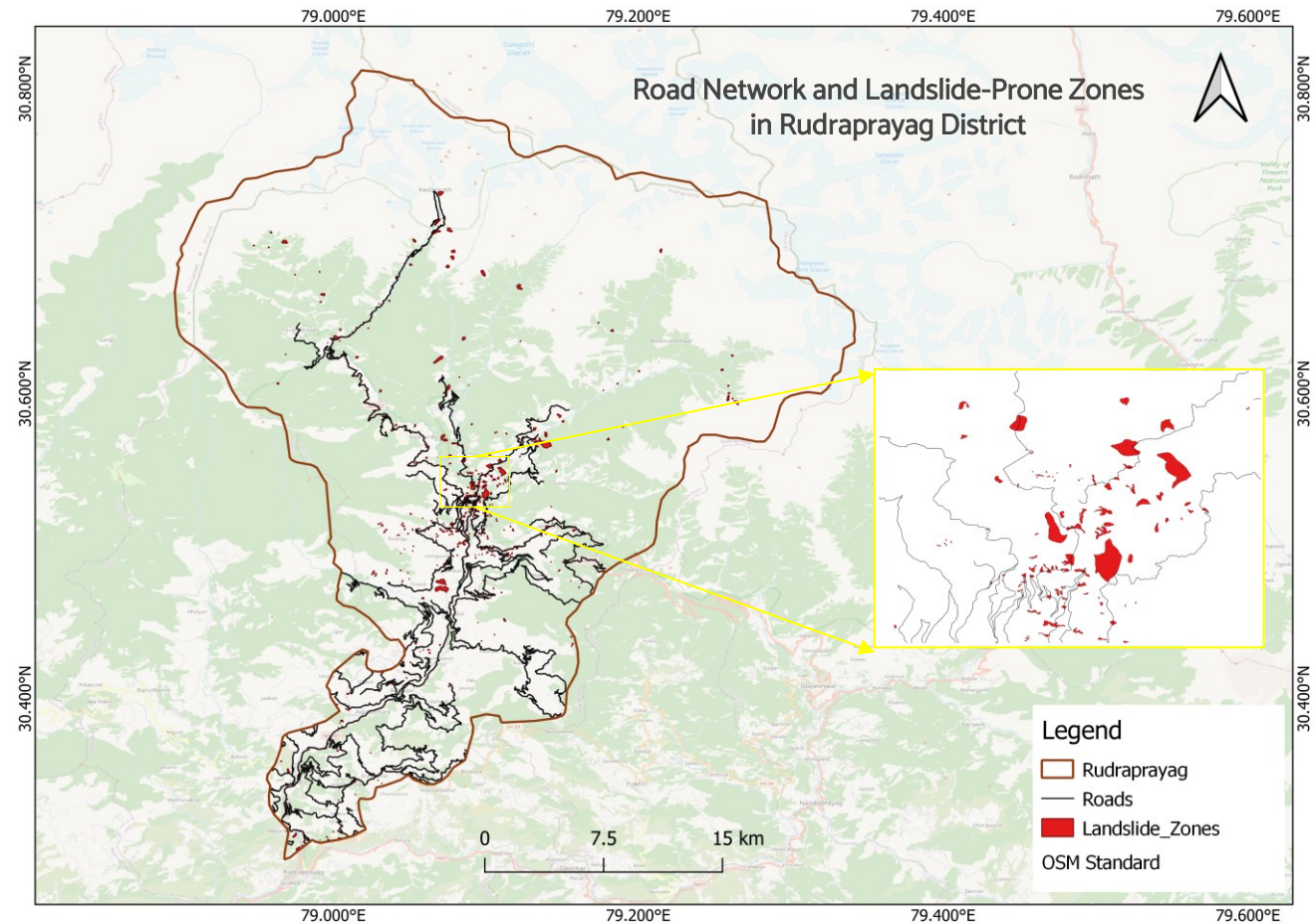
The terrain analysis demonstrates a strong correlation between **topographic parameters and landslide occurrence** in Rudraprayag. The slope map reveals extensive areas with **high slope gradients (up to 77°)**, indicating increased susceptibility to slope failure, while the hillshade map highlights the **complex terrain morphology, including steep ridges and deeply incised valleys**. Overlay analysis shows that landslide zones are predominantly concentrated in areas of **high slope and geomorphological complexity**, confirming that slope gradient and terrain variability are critical controlling factors in landslide susceptibility assessment.



Source:
Landslide zones: Geological Survey of India, DEM : SRTM 30m, USGS Earth Explorer



Landslide & Road Network Overlay Analysis

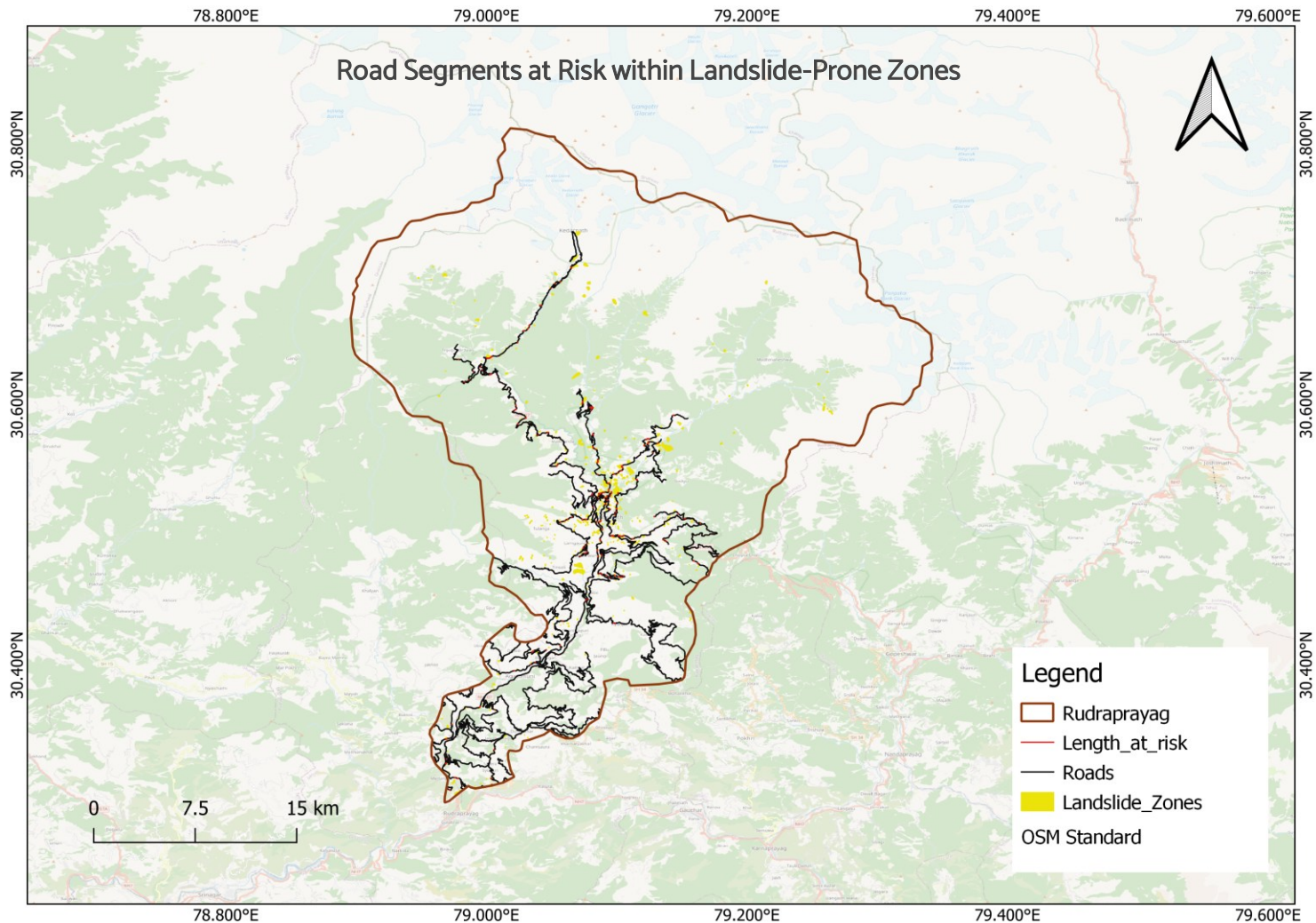


The overlay analysis reveals a significant spatial intersection between **road networks and landslide-prone zones** in Rudraprayag district. Several critical road segments pass through or lie in close proximity to identified landslide polygons, indicating a high level of **infrastructure exposure to slope instability**.

The concentration of affected road segments in specific corridors suggests that **terrain-driven hazards directly influence transportation vulnerability**, particularly in steep and geologically sensitive regions. These findings highlight the presence of **high-risk zones where road connectivity is susceptible to disruption**, emphasizing the need for targeted monitoring, maintenance, and risk mitigation strategies.

Source:

Landslide zones: Geological Survey of India, Roads : OSM



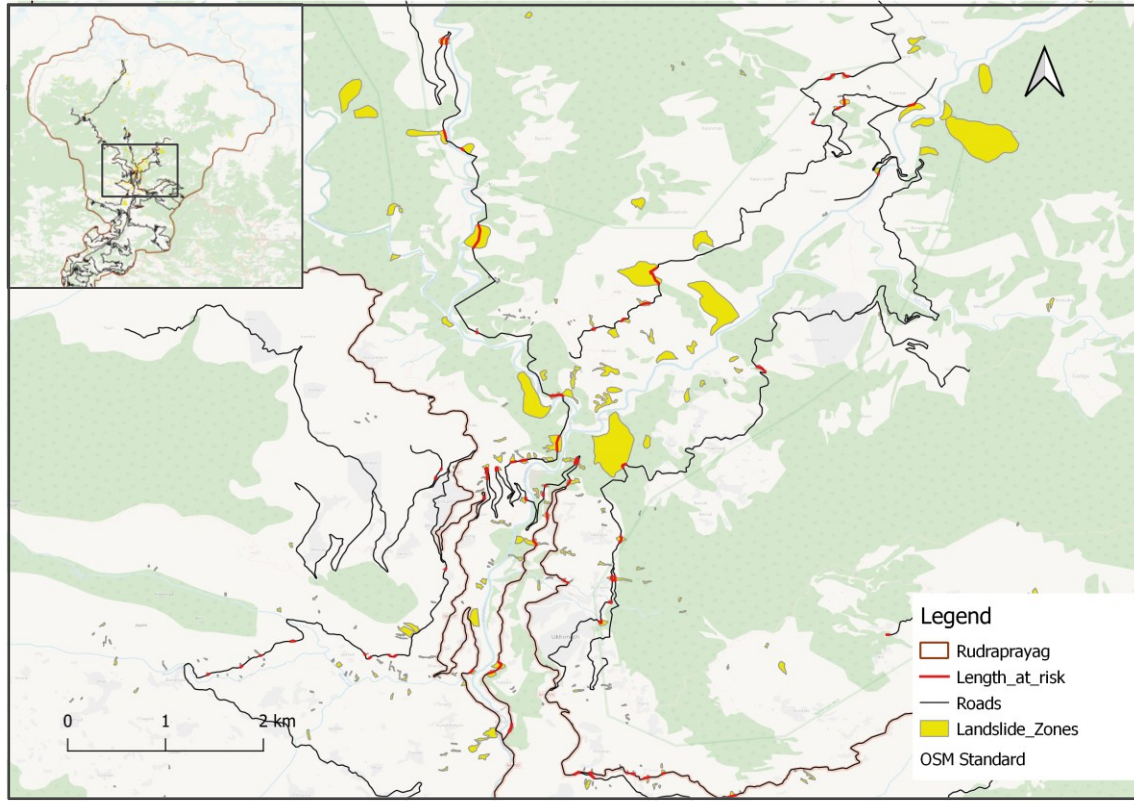
Inference:

The map illustrates the road segments at risk due to landslides, showing how much of the transportation network falls within landslide-prone areas. The road network was clipped to the landslide polygons, showing the sections exposed to hazard zones.

Source:

Landslide zones: Geological Survey of India
Roads : OSM

Road Exposure & Risk Quantification



Roads at Risk within Landslide Zones

Total Road Length at Risk: 7,528.39 meters

Statistics of length_risk

Field

Length

Statistics:

Count: 79

Minimum: 0.84

Maximum: 708.43

Sum: 7528.39

The quantitative analysis indicates that a total of **7,528.39 meters of road network** falls within **landslide-prone zones**, highlighting a significant extent of infrastructure exposed to potential hazards.

These vulnerable segments are concentrated in specific **high-risk corridors**, where slope instability poses a direct threat to connectivity and accessibility. This assessment provides a critical basis for prioritizing risk mitigation and infrastructure resilience planning.

Conclusion & Recommendations

Key Findings

- Rudraprayag district exhibits high landslide susceptibility due to steep slopes and complex terrain conditions.
- Terrain analysis confirms that slope gradient and elevation variability are major controlling factors for landslide occurrence.
- Overlay analysis shows that several road segments intersect landslide-prone zones, indicating high infrastructure vulnerability.
- Approximately 7,528.39 meters of road network is exposed to landslide hazards.
- High-risk corridors are concentrated in steep and geologically sensitive regions, affecting connectivity and safety.

Recommendations

• Slope Stabilization Measures

Implement retaining walls, rock bolting, and slope reinforcement techniques in identified high-risk zones.

• Improved Drainage Management

Develop proper surface and subsurface drainage systems to reduce water infiltration and slope failure.

• Land Use Regulation

Restrict construction and excavation activities in **high landslide susceptibility zones** through zoning policies.

• Road Design Improvements

Realign vulnerable road segments and adopt **slope-sensitive engineering designs** to minimize hazard exposure.

• Regular Monitoring & Early Warning Systems

Install sensors and monitoring systems in critical zones for **real-time landslide detection and alerts**.

• Afforestation & Bioengineering Techniques

Promote vegetation cover to enhance slope stability and reduce soil erosion.